Combined effect of External Treatment of Herbal Medicine and Tuina in Congenital Muscular Torticollis: Systematic Review and Meta-analysis

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Supplement File 2. Tuina treatment information

First author (year)	Target muscles or acupoints	Manipulations	Tuina massage medium (E/C)	Treatment duration	Additional education
Cai (2014)	SCM muscle	Pushing, kneading (10 minutes): on the mass of the SCM muscle Flicking, plucking (3 minutes): on the mass of the SCM muscle Pulling, rotating: affected shoulder and neck (stretch the affected side of the SCM muscle, head is rotated to the affected side) (3~5 times)	talcum powder/ talcum powder	15~20 min/time, 3 times/w	NR
Dai (2021)	SCM muscle, posterior neck and shoulder muscle Gyeonjeong (GB21)	Grasping (3 minutes): on the mass of the SCM muscle (100~120 times/min) Pressing, kneading flicking, plucking (10 minutes): (100~120 times/min) Passive stretching (15 minutes): affected shoulder and neck (stretch the affected side of the SCM muscle (20~30 times/min) Pressing, kneading (3 minutes): the SCM muscle, posterior neck and shoulder muscle, GB21	Johnson & Johnson baby Powder/Johnson & Johnson baby Powder	1 time/d, 5 times/w	NR
He (2009)	SCM muscle, splenius capitis muscle origin, trapezius muscle, face and mandible region	Pressing, kneading (5~15 minutes): on the mass of the SCM muscle Pinching, Pressing, kneading, rubbing (5~10 minutes): the splenius capitis muscle origin(nuchal ligament), trapezius muscle, face and mandible region Rotating: (Suitable for more than 2 months child) head is rotated and lifted up slightly (20~30 times) Stretching: shoulder and neck (10~20 times) Pulling, rotating: rotate the patient's face forward and obliquely upward by 45° (3~5 times)	NR	5 times /w	E: Children at 3 months of age may perform prostrate head-up and neck exercises
Li (2018)	SCM muscle, trapezius muscle, the posterior cervical muscle group, Inyeong (ST9), Sudol (ST10), Budol (LI18), Gyeonjeong (GB21), Pungji (GB20)	Pressing, kneading: the SCM muscle, the upper third part of the trapezius muscle and the posterior cervical muscle group, ST9, ST10, LI18, GB21, GB20 Pulling: head is pulled to the unaffected side Pressing, kneading: the SCM muscle and shoulder muscle group	NR	30 min/time, 1 time/d, 6 times /w	NR
Ren (2008)	SCM muscle, neck and shoulder muscles	Kneading: neck and shoulder muscles Pulling: the SCM muscle on the affected side(5~10 times) Kneading (2 minutes): (if the face is deformed or asymmetrical) the face	NR	15~20 min/time, 1 time/d	NR
Tian (2010)	SCM muscle, bilateral shoulder and back muscles	Pressing, rubbing, kneading, grasping (5~8 minutes): the SCM muscle on the affected side Pinching, flicking, plucking, pulling, stretching (3~5 minutes): affected side of the SCM muscle Pulling, rotating, passive stretching (10~20 seconds): head is pulled up and rotated toward the unaffected, lengthen the SCM muscle tendon on the affected side (4~6 times) Kneading, pinching: the entire SCM muscle and bilateral shoulder and back muscles	NR	20~30 min/time, 1 time/d	correct position of the head and neck, place a clean cotton pillow on both sides of the head

Wang (2008)	SCM muscle	Pressing, rubbing, flicking, plucking: the SCM muscle on the affected side Pulling, rotating: shoulder and neck	ETHM (self-made stretch liquid)/ talcum powder	1 time/d	NR
Wang (2011)	SCM muscle, trapezius muscle, Gyeonjeong (GB21), Pungji (GB20)	(1) mass type kneading (5 minutes): the whole SCM muscle on the affected side grasping: the SCM muscle on the affected side (1~2 times) stretching: head is tilted to the shoulder of the unaffected side (more than 10 times) kneading, pinching: the trapezius muscle on the affected side, GB21, GB20 (2) not mass type grasping: the SCM muscle on the affected side (2~3 times) pushing, passive stretching: head is tilted to the shoulder of the unaffected side, passive lateral movements in the frontal plane rubbing, Kneading (5 minutes): the SCM muscle on the affected side	talcum powder/ talcum powder	20~30min/time, 1 time/d	NR
Wang (2016)	SCM muscle, neck and shoulder muscles	Kneading (5~10 minutes): the bilateral neck and shoulders Pulling (2~5 minutes): stretch the SCM muscle on the affected side Rotating (2 minutes): head is rotated (5~10 times)	talcum powder/ talcum powder	15~20min/time, 1 time/d	NR
Wang (2018)	SCM muscle, Qiaogong point(the connection between Yepung(TE17) and Gyeolbun(ST12)),Gyeonjeong (GB21)	Pressing, kneading: the SCM muscle on the affected side and Qiaogong point Flicking, plucking (10 minutes): the SCM muscle on the affected side (starting point, end point and mass) (100~120 times/min) Pinching (2 minutes): the SCM muscle on the affected side (100~120 times/min) Passive stretching, rotating(15 minutes): head is passively rotated towards the affected side, the affected SCM muscle is elongated (20~30 times) Grasping, pressing, kneading (2 minutes): GB21, the trapezius muscle and neck	Johnson & Johnson baby Powder/ Johnson & Johnson baby Powder	1 time/d, 5 times /w	NR
Wang (2019)	SCM muscle, the muscles of the spine	Kneading: the SCM muscle on the affected side Pushing: the bilateral SCM muscle Pulling: head is pulled to the unaffected side Rubbing: the bilateral the muscles of the spine	talcum powder /talcum powder	20 min/time, 1 time/d	NR
Xu (2017)	SCM muscle	Kneading, pushing (5 minutes): the SCM muscle on the affected side (100~120 times/min) Pinching, grasping, plucking (5 minutes): the SCM muscle on the affected side (100~120 times/min) Pulling: head is curvatured towards the unaffected side (20 times for a cycle, repeated 4 cycles) Rotating: head is rotated towards the affected side (20 times for a cycle, repeated 4 cycles)	ETHM (Bai Mai ointment)/ lubricating oil	20 min/time, 1 time/d, 5 times/w	NR
Xu (2018)	SCM muscle	Kneading, pinching (7 minutes): the SCM muscle on the affected side Pulling: head is pulled toward the unaffected shoulder (20~30 times) Rotating: the mandible is rotated toward the shoulder of the affected side (20~30 times)	ETHM (Sanse san ointment) / talcum powder	1 time/d	NR
Yi (2020)	SCM muscle, neck and shoulder muscles of the affected side	(1) mass type rubbing, pressing, kneading, flicking, pinching, pulling: the SCM muscle and the neck and shoulder muscles of the affected side (2) not mass type	NR	20~30 min/time, 1 time/d, 6 times/w	NR

		rubbing, pressing, kneading, pinching, pulling: the SCM muscle and the neck and shoulder muscles of the affected side			
Yuan (2021)	SCM muscle	Pressing, kneading (5 minutes): the mass of the SCM muscle on the affected side Grasping: the mass of the SCM muscle on the affected side (3~5 times) Pushing, kneading, flicking, plucking: the tendons of the SCM muscle on the affected side (10~15 times) Passive stretching: increase the distance between the ear and the shoulder of the affected side (20 times) Rotating, pulling: the mandible is turned to the affected side and the head is tilted to the healthy side (10 times) Pressing, kneading (3~5 minutes): the mass of the SCM muscle on the affected side	ETHM (Sanse san ointment) / talcum powder	1 time/d, treatment for 10 d and rest 2 d	NR
Zhang (2012)	SCM muscle, Qiaogong point,trapezius and supraspinatus	Kneading (5 minutes): the SCM muscle on the affected side Pulling (2 minutes): head is pulled toward the unaffected side Rotating (2 minutes): head is rotated Rubbing (5 minutes): Qiaogong point Pinching, kneading (6 minutes): trapezius and supraspinatus on the affected side	talcum powder/ talcum powder	20 min/time, 1 time/d	NR
Zhang (2019)	Cheonjong(SI11), Gyeonjeong (GB21), Pungji (GB20), Budol (LI18) and other related acupoints	Grasping, pressing, kneading, flicking, plucking, pulling: SI11, GB21, GB20, LI18 and other related acupoints	talcum powder/ talcum powder	20~30 min/time, 1 time/d	NR
Zhang (2020)	Pungji (GB20), Inyeong (ST9), Budol (LI18), Yepung (TE17), Gyeonjeong (GB21) and Ashi points	Rubbing: face and neck Pressing, kneading: GB20, ST9, LI18, TE17, GB21 and Ashi points Flicking, plucking: the mass of the SCM muscle on the affected side Pinching: the mass of the SCM muscle on the affected side Pulling, rotating: head is pulled and rotated to the maximum angle on both sides (3~4 times)	NR	NR	NR
(Zhao 2016)	SCM muscle, neck and trapezius muscle, occipital bone	Kneading (5 minutes): occipital bone, neck, and trapezius muscle on the affected side Kneading (5~8 minutes): the mass of the SCM muscle on the affected side Grasping, pinching, Kneading (3 minutes): the SCM muscle on the affected side Stretching: head is inclined slowly towards the shoulder of the unaffected side Kneading, Pulling: midpoint of the SCM muscle (10 times)	NR	20 min/time, 1 time/d	NR

* SCM = Sternocleidomastoid, E = experimental group, C = control group, GB = Gallbladder Meridian, ST = Stomach Meridian, LI = Large Intestine Meridian, TE = Triple Energizer Meridian, SI = Small Intestine Meridian, EHTM = external treatment of herbal medicine, NR = not reported, min = minute, d = day, w = week, m = month